

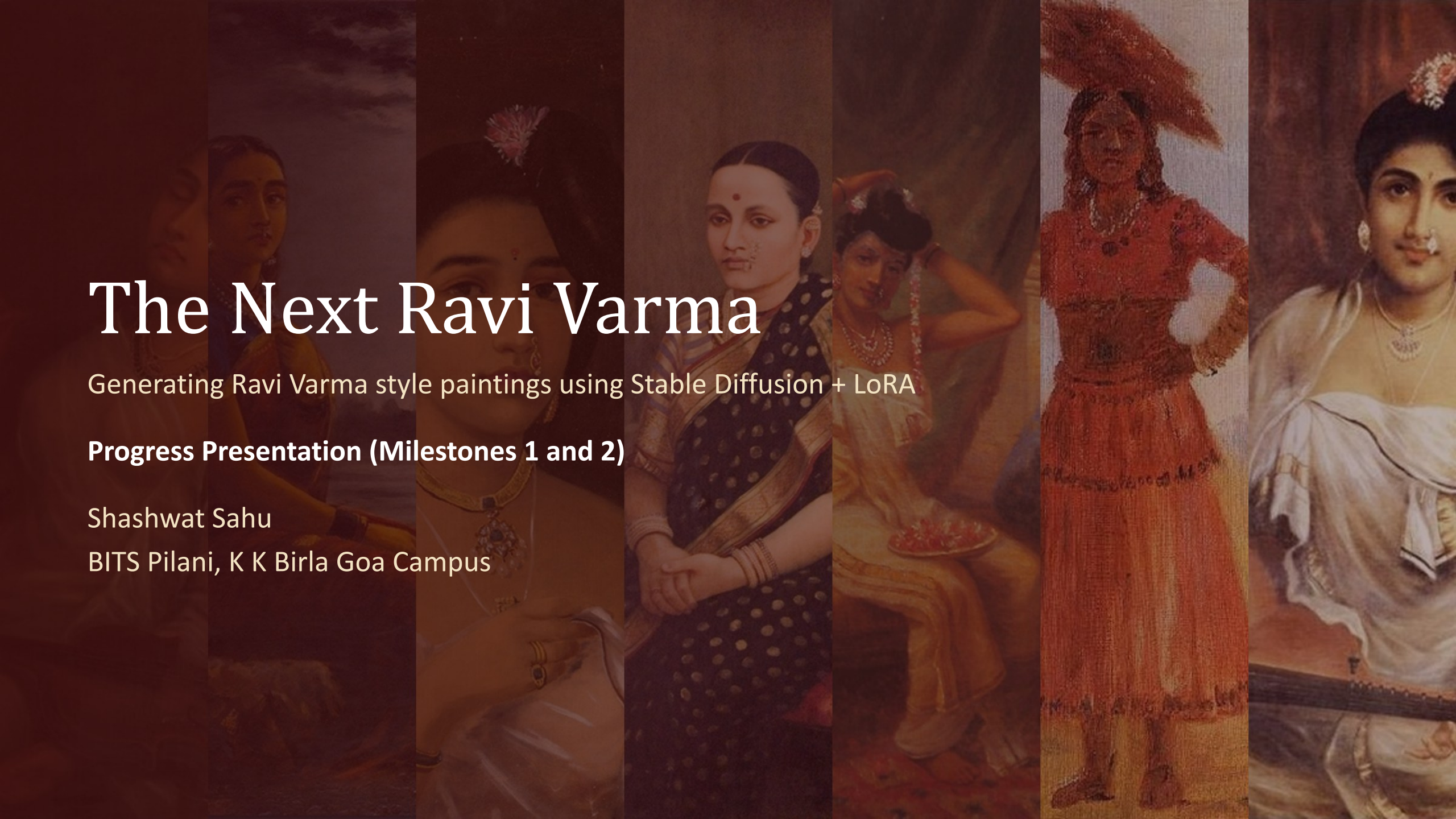
The Next Ravi Varma

Generating Ravi Varma style paintings using Stable Diffusion + LoRA

Progress Presentation (Milestones 1 and 2)

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Introduction

- Raja Ravi Varma (1848-1906) was a famous Indian painter known for realistic oil paintings of Indian mythology and royal portraits.
- Inspired by "The Next Rembrandt" project, which generated a new painting in Rembrandt's style.
- We use Stable Diffusion 1.5 and fine-tune it with LoRA on Ravi Varma's paintings.

Our goal: give a text description of a scene and generate a new image in Ravi Varma's style.



Hamsa Damayanti by Raja Ravi Varma

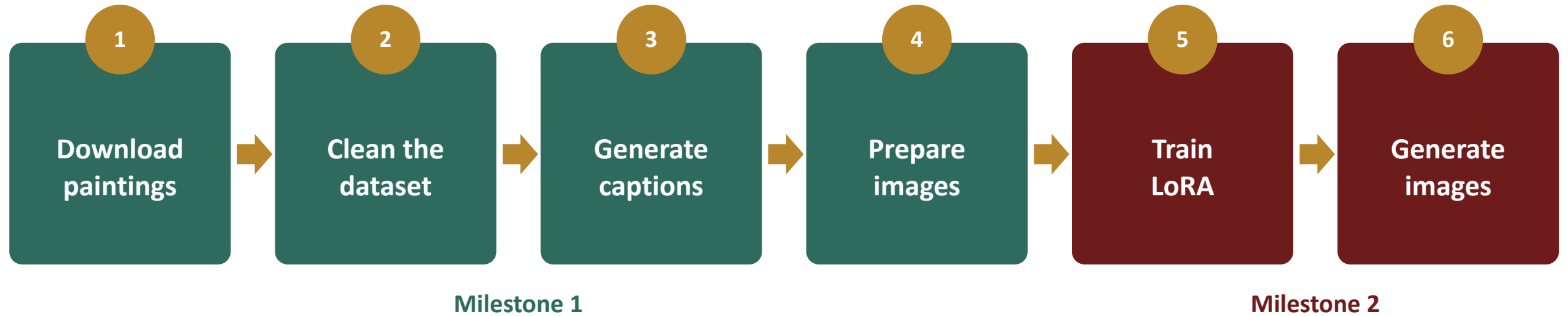
Project Milestones and Status

1	Dataset collection, cleaning and captioning	Done
2	LoRA fine-tuning of Stable Diffusion	Done
3	Prompt expansion using an LLM	Not started
4	ControlNet (pose control), evaluation, web UI	Not started



2 of 4 milestones completed

Overall Pipeline



- Steps 1-4 run on my laptop (MacBook), training runs on Google Colab (T4 GPU).
- Everything is controlled by config files and scripts, so the results can be reproduced.

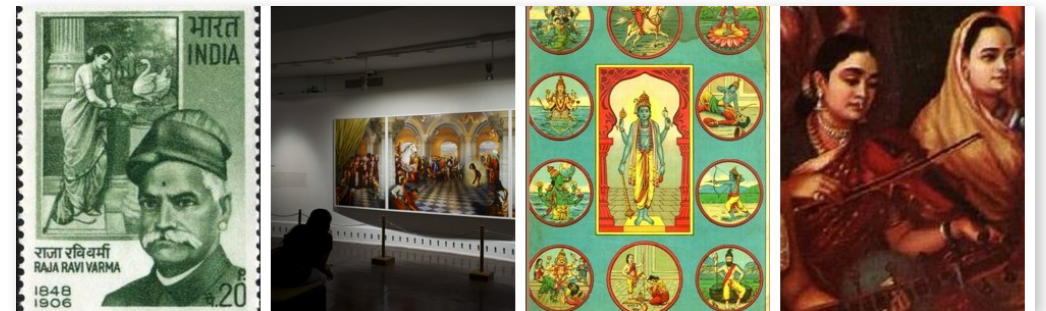
Milestone 1: Dataset



- Downloaded from Wikimedia Commons (most are public domain).
- Removed 26 low-resolution, 33 duplicate and 7 non-painting images.
- 12 images kept aside for testing.
- Images resized without cutting off heads/feet.



Some training images



Examples of removed images

Milestone 1: Image Captioning



- Each image needs a text caption for training.
- Used BLIP (an image captioning model) to describe the image and BLIP-VQA to answer fixed questions (clothing, jewellery, pose, setting, etc.).
- Wrong answers like "phone" were filtered out.
- Every caption starts with a trigger word <rvvarma> so the model links it to the style.

Example caption

<rvvarma>, Hamsa Damayanti, a woman in a red sari feeding a white swan, wearing red dress, necklace jewelry, sitting, calm expression ...

Milestone 2: LoRA Training

3.2M

parameters trained
(0.37% of the model)

~1 hr

on Google Colab
(T4 GPU)

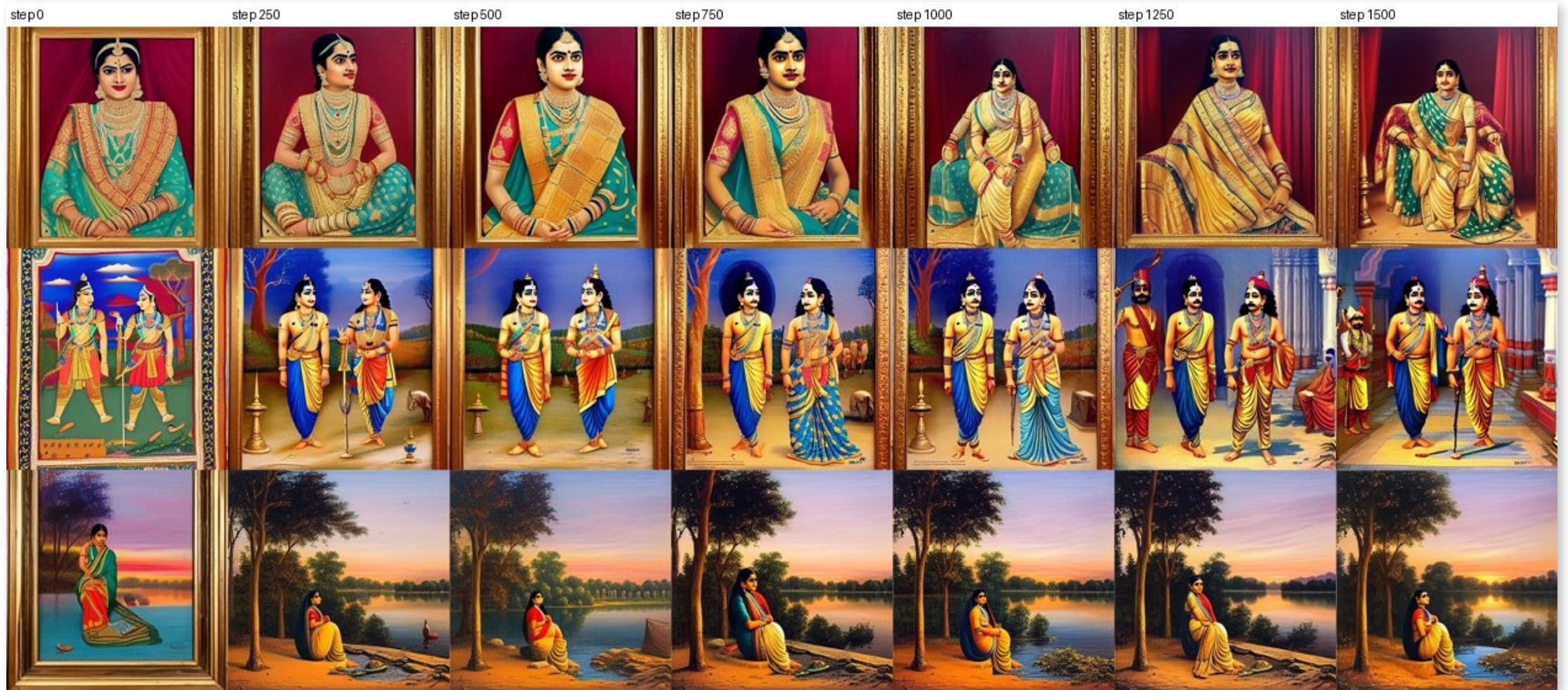
1500

training steps

- LoRA trains only a small number of extra weights; the original Stable Diffusion model is not changed.
- Saved a checkpoint every 250 steps and generated test images to track progress.

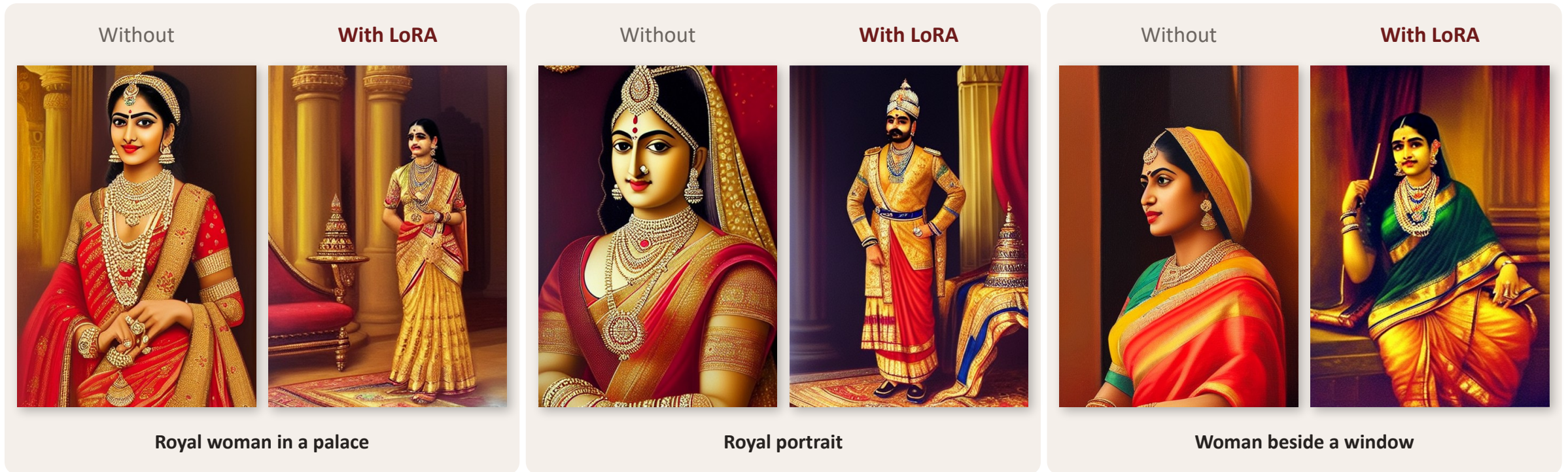
Parameter	Value
Base model	Stable Diffusion 1.5
LoRA rank / alpha	16 / 16
Learning rate	0.0001
Batch size	1 (gradient accumulation 4)
Training steps	1500
Precision	fp16

How the Output Changed During Training



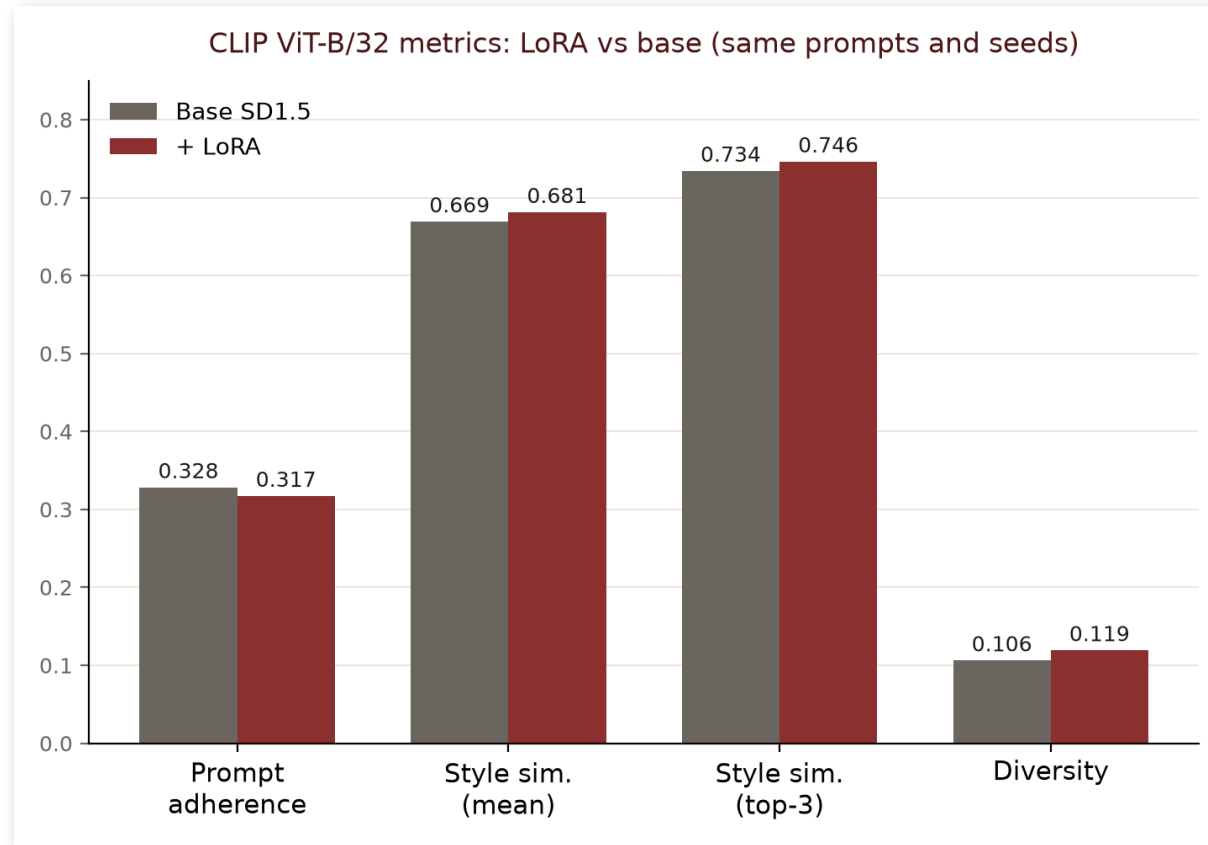
Same prompts and seeds, from step 0 (no training) to step 1500

Results: Without vs With Our LoRA



- With LoRA the images look more like Ravi Varma's style: full-figure poses, pillars, carpets, darker painted backgrounds.
- The base model added picture frames in 5 of 15 images; our model added none.

Evaluation and Problems



CLIP scores for 15 image pairs

What we found

- CLIP style score increased only slightly (0.669 to 0.681), even though the images look clearly different.
- No image was a copy of a training image.

Problems

- Some women's faces get a dark shadow above the lip.
- Small signature-like marks appear in the corners.
- Group scenes still look like posters.

Challenges Faced



Many images in the dataset were duplicates or had been upscaled, so we had to check every image.



The original Stable Diffusion 1.5 model was removed from Hugging Face, so we switched to the official mirror.



Version conflicts on Google Colab (xformers, bitsandbytes) broke training at first.



On the Mac, images came out completely black because of an fp16 issue; fixed by changing a setting.



Free Colab has limited GPU time, so we added checkpoints to resume training if it disconnects.

Future Work

1 Fix the face shadow and signature problems by improving the captions.

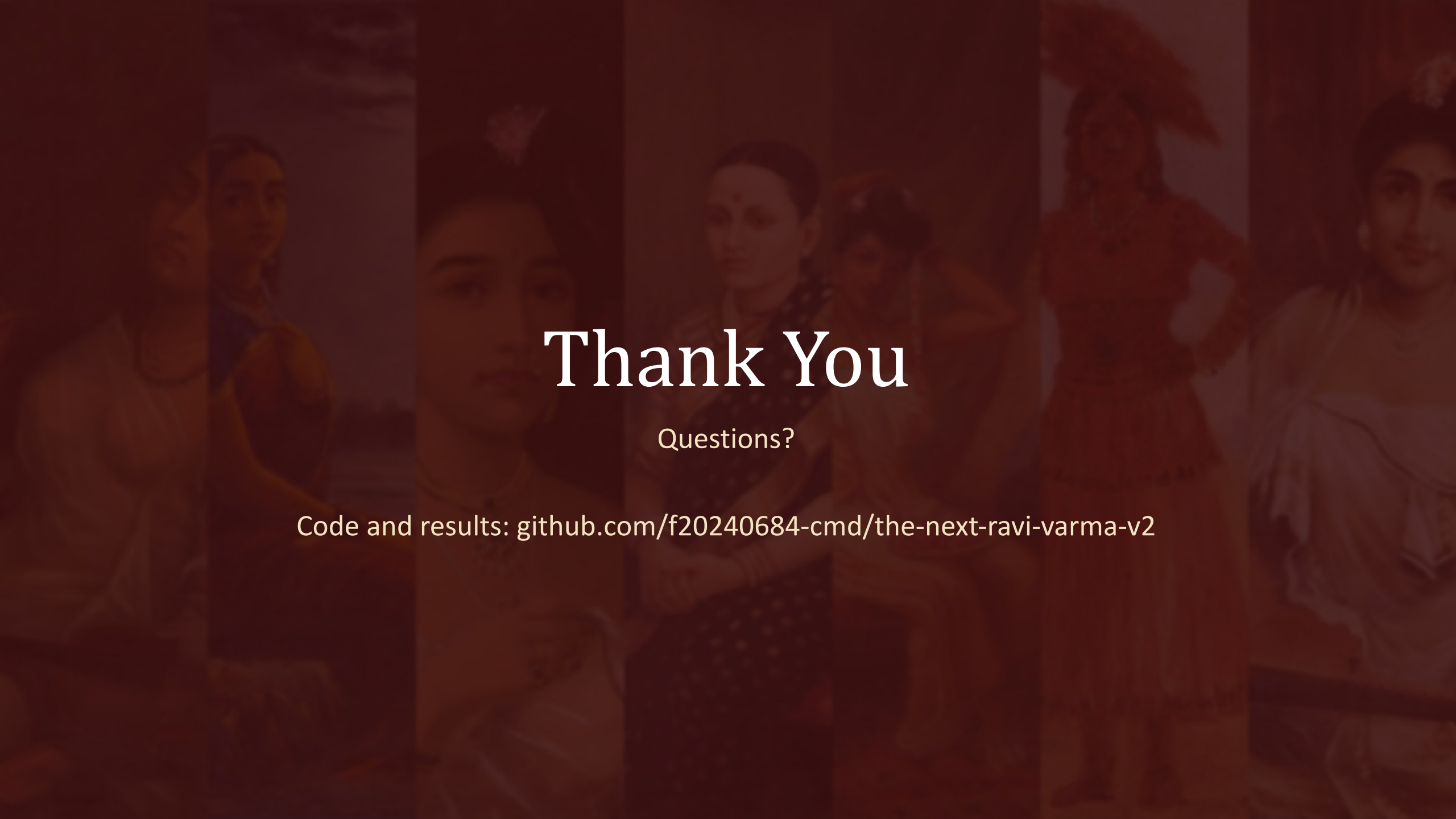
2 Milestone 3: use an LLM to turn a story description into a detailed prompt (e.g. "Krishna and Arjuna at Kurukshetra").

3 Milestone 4: add ControlNet so the user can control the pose of the people.

4 Build a simple web interface (Gradio) and do a better evaluation, including human ratings.



Current output for a Krishna and Arjuna prompt



Thank You

Questions?

Code and results: github.com/f20240684-cmd/the-next-ravi-varma-v2